MTH2008

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2021

Real Analysis

Module Leader: Dr Mark Callaway

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

SECTION A

1. (a) Consider the following subset of the real numbers:

$$S = (-1, 2) \cup [3, \infty).$$

(i) Is the set S open? Is the set S closed? In each case justify your answer.

(ii) Determine the interior of S and the set of boundary points of S. (8)

(b) Prove, using an "ϵ-δ" argument, that the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \dfrac{1}{x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is not continuous at $x_0 = 0$. (10)

(c) Let $\{s_n\}$ be the sequence

$$s_n = \sin\left(\frac{n\pi}{2}\right) + \cos(n\pi).$$

Find all numbers L in the extended reals that are limits of some subsequence of $\{s_n\}$, and, for each such L choose a subsequence $\{s_{n_k}\}$ such that $\lim_{k\to\infty} s_{n_k} = L$. Does the sequence $\{s_n\}$ converge? Justify your answer. (8)

(d) Let $\{f_n\}$ be the sequence of functions $f_n : [0, 1] \to \mathbb{R}$ defined by

$$f_n(x) = \frac{x^n}{n}$$

and let $f(x) = \lim_{n\to\infty} f_n(x)$, where the limit exists. Find f. Does $\{f_n\}$ converge uniformly to f? Justify your answer. (12)

(e) Show that the function $f : \mathbb{R}^3 \to \mathbb{R}$ given by

$$f(x) = x_1^3 + x_1 x_3 + x_2^2 - 2x_2$$

for all $x = (x_1, x_2, x_3)^\intercal \in \mathbb{R}^3$ is differentiable and find the derivative. Hence determine the stationary points of f. (12)

[**50**]

SECTION B

2. (a) Prove or disprove (using a counterexample) the following statements.

   (i) A limit point of a set S is either an interior point of S or a boundary point of S.

   (ii) A set S has no limit points if and only if each of its points is isolated.

   (iii) The boundary ∂S of a set S is closed. (15)

   (b) Suppose that $f : [a, b] \to \mathbb{R}$ is continuous and that $f(x)$ is irrational for all $x \in [a, b]$. Prove that f must be a constant function. (10)

   [25]

3. (a) Assume that $f'(x) > g'(x)$ for all $x \in \mathbb{R}$ and that $f(x_0) = g(x_0)$ for some $x_0 \in \mathbb{R}$.

   (i) Show that $f(x) > g(x)$ for $x > x_0$ and $f(x) < g(x)$ for $x < x_0$. (*Hint: consider the function* $f - g$.)

   (ii) Show, by providing an example for each case, that if we omit the assumption that $f(x_0) = g(x_0)$ for some $x_0 \in \mathbb{R}$, then it is possible that $f(x) > g(x)$ for all $x \in \mathbb{R}$ or that $f(x) < g(x)$ for all $x \in \mathbb{R}$. (10)

   (b) Prove that if f is a continuous non-negative function on $[a, b]$ and $\int_a^b f(x)\,dx = 0$, then $f(x) = 0$ for all $x \in [a, b]$. (*Hint: try a proof by contradiction.*) Show by an example that the conclusion doesn't hold if we omit the assumption that f is continuous. (15)

   [25]

4. (a) Prove that the function $f : [0, 1] \to \mathbb{R}$ where

$$f(x) = \sqrt{x}$$

is Riemann integrable on $[0, 1]$. (10)

(b) Find the derivative of the function $f : [0, \infty) \to \mathbb{R}$ given by

$$f(x) = \cos\left(\int_0^{\sin(x)} e^{-t^2}\, dt\right)$$

justifying your steps. (5)

(c) Prove, using an "ϵ-δ" argument, that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \dfrac{2x_1x_2}{\sqrt{x_1^2 + x_2^2}}, & x \neq (0, 0)^\intercal \\ 0, & x = (0, 0)^\intercal \end{cases}$$

for all $x = (x_1, x_2)^\intercal \in \mathbb{R}^2$ is continuous at $x = (0, 0)^\intercal$. *Hint: you may like to make use of Young's inequality* $|ab| \leq \frac{1}{2}(a^2 + b^2)$. (10)

[25]